Progressive Clarity with Honest Constraint:

A Design Framework for Hormone Replacement Therapy Tracking in the xT Mobile Application

Starla Cortes

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Abstract

We present the user experience architecture of xT, a mobile application that predicts serum hormone concentration over time from a user's logged dose history for gender-affirming hormone replacement therapy (HRT). The application translates a pharmacokinetic (PK) model spanning ten medication/route/form combinations into a consumer interface designed for daily use by people without medical training.

The UX framework rests on three principles: **progressive disclosure**, which reveals complexity only as the user demonstrates readiness—from an 11-step onboarding wizard through to a 4-tab daily experience with tenure-tiered content that unlocks over months and years of use; **honest presentation**, which reports hormone levels as population estimates rather than facts, gates comparative judgments behind user-configured therapeutic targets, and phrases safety flags as clinical associations rather than directives; and **per-user personalization**, which adapts the interface to the individual through onboarding choices, blood test calibration, body-type selection, and injection-zone preferences.

We describe the full user journey from first launch through ongoing mastery, document the design-token system that encodes clinical semantics as visual properties, validate the architecture against eight persona journeys that exercise UX edge cases, and provide an honest accounting of design trade-offs: where the UX is anchored, where it is reasoned but untested, and where it remains placeholder. The paper is written to be checkable: every design decision is cited to its source in the codebase, and every limitation is named.

Keywords: user experience, mobile health, progressive disclosure, hormone replacement therapy, design system, onboarding, personalization, pharmacokinetic visualization

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1 Introduction

Gender-affirming hormone replacement therapy (HRT) involves the administration of exogenous hormones—estradiol and/or testosterone—to align an individual's serum hormone levels with their gender identity. Clinical management relies on periodic blood draws to measure serum hormone concentration, titrate dosage, and confirm therapeutic levels [2]. Between draws, patients have limited tools to anticipate where their levels are heading, when a peak or trough is expected, or how a missed dose or regimen change will affect the trajectory.

xT addresses this gap with a pharmacokinetic model that predicts serum hormone concentration over time from the user's logged dose history. The model itself is described in a companion paper [1]; this paper concerns the *interface* that delivers that model to its users. The challenge is non-trivial: the underlying mathematics are undergraduate-level differential equations, the clinical context is high-stakes, and the audience is anyone on HRT—not just those comfortable with exponential decay curves.

1.1 The UX Problem

People on HRT must manage a constellation of tasks: tracking injection schedules across dosing intervals that range from daily to quarterly, rotating injection sites across body zones to prevent tissue damage, interpreting hormone level estimates in the context of therapeutic ranges, correlating subjective experience (mood, symptoms) with objective data (blood test results), and anticipating physical changes over months and years. None of these tasks require medical training, but all of them benefit from clinical context.

Most available tools fall into one of two categories. General-purpose medication reminders reduce the problem to a binary "did you take it" checkbox, discarding the pharmacokinetic information that makes sense of *when* a level is measured and *where* in the cycle the user sits. Clinical reference materials, at the other extreme, present PK curves and population tables without translation—accurate but illegible to someone who has never seen a biexponential.

xT occupies the space between: it computes a full pharmacokinetic prediction but presents it as a single number, a single curve, and a single status. The complexity is present but gated; detail is available on demand but never imposed.

1.2 The Design Response

xT's interface is organized around three principles that operate at every level of the application, from the 11-step onboarding wizard to the ambient constellation visualizations on the Time tab.

Progressive disclosure. The app never shows more than the user needs at their current stage. Onboarding introduces one concept per step. The home screen foregrounds exactly one number. The Time tab's content is gated by tenure tier—a user in month 1 does not see year-3 surveillance content. Detail lives behind taps, and taps are only offered when the detail is relevant.

Honest presentation. The PK model produces population estimates, not individual predictions. The interface reflects this at every level: level readouts are labeled "Estimated," comparative judgments ("below target") only appear against user-configured therapeutic targets, and safety flags state clinical associations ("Estradiol above 1,000 pg/mL is associated with increased thromboembolic risk") rather than directives ("You should lower your dose"). The app reports what the model says; it does not advise.

Per-user personalization. Between-subject variability in hormone metabolism is substantial—published studies report coefficients of variation of 0.58 on V_d/F for injectable testosterone [3], corresponding to an ∼8-fold individual range. The interface addresses this through a chain of personalization: medication type and ester at onboarding, a per-user calibration factor α derived from the user's own blood tests, body-type selection for the 3D injection-site model, and user-configurable therapeutic target bands. No single setting carries the full weight of individualization; each layer tightens the fit.

1.3 Contributions

1. A complete walkthrough of the UX architecture, from 11-step onboarding through the 4-tab daily experience, with design rationale for each structural decision.
2. A design-token system that encodes clinical semantics (level status, safety flags, cycle position, tenure tier) as visual properties without presenting medical advice, implemented in a single presentation-rules file (`HormoneLevelPresentation.swift`) that separates semantics from display.
3. A progressive disclosure framework that operates on three timescales simultaneously: the onboarding sequence (minutes), the per-tab depth gradient (seconds per interaction), and the tenure-tier system (months to years).
4. A per-user calibration interface that translates a blood test result into a single multiplicative factor α, absorbing the dominant source of individual-level prediction error without requiring the user to understand population pharmacokinetics.
5. An honest accounting of UX limitations—what is designed well, what is reasoned but untested, and what remains unaddressed—in the tradition of the companion PK paper's explicit marking of calibrated versus uncalibrated parameters.

2 Design Principles

This section describes the five principles that govern every design decision in xT. They are the UX analog of the PK model's governing equation (Equation 1 in [1]): every screen, every animation, and every string of user-facing text is answerable to at least one of them.

2.1 State Facts, Don't Advise

The central tension in any health-tracking application is the line between *informing* and *advising*. xT's position is that the line exists, that crossing it requires clinical authority the app does not possess, and that the interface must be designed to stay on the correct side of it.

This principle is implemented in `HormoneLevelPresentation.swift`, the single file through which every hormone level estimate becomes user-facing text. The file exports classification enums (`LevelStatus`, `SafetyFlag`, `ModelDeviation`, `RangeSource`) and a set of formatting functions. Its design follows three rules:

1. **Classifications are values, not display strings.** The `LevelStatus` enum has cases `criticallyLow`, `veryLow`, `slightlyBelow`, `inRange`, `slightlyAbove`, and `high`—but these are model facts, not verdicts. Views build their own text from these values, which keeps semantics and presentation (and therefore localization) separate. The app never ships a hardcoded English string like "Your levels are low" from the model layer.
2. **Comparisons are gated on user-set bounds.** The field `RangeSource` distinguishes `userSet` bounds (the user has configured therapeutic targets in Settings) from `modelDerived` bounds (population defaults). Comparative copy—"within your set trough range," "below your set target"—is only rendered when the source is `userSet`. If the user has not configured targets, the app reports the estimate without judgment. The model may know that 80 pg/mL is below the typical estradiol trough, but the app does not tell the user they are "below target" against a target they never set.
3. **Safety flags state associations, not instructions.** The `SafetyFlag` enum identifies crossings of documented clinical thresholds: `estrogenCeiling` (E2 $>$ 1000 pg/mL), `testosteroneSupraphysiologic` (T $>$ 1500 ng/dL), `subTherapeuticTrough`, and `spironolactoneClearance`. The user-facing text for each flag states the association ("Estradiol above 1,000 pg/mL is associated with increased thromboembolic risk") without prescribing action. The app's position is that stating the association is informing; saying "you should lower your dose" is advising, and the app does not do it.

This principle has a cost: it makes the interface less directive than some users might prefer. A user who wants the app to say "take your dose now" or "your levels are too high—skip today" will find xT reticent. That reticence is intentional. The companion PK paper [1] notes that the model's predictions "are population estimates that should not be used to make dosing decisions without laboratory confirmation"; the interface honors that constraint by refusing to translate population estimates into individual instructions.

2.2 Progressive Disclosure

xT practices progressive disclosure on three timescales simultaneously.

Onboarding (minutes). The 11-step onboarding wizard (Section 3.1) introduces exactly one concept per screen: first the hormone type, then the experience level, then the administration method, and only then the ester—a term many users encounter for the first time in the app. Each step builds on the selections made in prior steps: the ester picker only appears after "Injectable" is chosen; the concentration picker is pre-populated with common values for the selected ester; the injection-site step is skipped entirely for oral routes. The user is never asked a question whose context they have not already been given.

Daily use (seconds per interaction). The home screen foregrounds exactly one number: the current estimated hormone level, displayed in large type with units and a status descriptor. The full PK curve is present as a background layer—visible but ambient, not demanding attention. The glass card stack below it contains four rows, each a single piece of information: insight text, today's mood, next dose countdown, and action buttons. Tapping any row opens detail; until tapped, the row is a summary. The design principle is that the home screen should be legible in under two seconds and actionable in under five.

Tenure (months to years). The Time tab's content is gated by `TenureTier`, a 7-level progression from `beginning` (0–2 months) through `veteran` (120+ months). Each tier unlocks specific content about expected changes: a user in month 1 sees information about initial physical responses; a user at year 3 sees content about long-term health surveillance. This gating is not gamification—the tiers are not displayed as achievements to chase—but a content-management strategy: a user in their first month does not need to read about year-3 mammogram schedules, and presenting that information prematurely would be noise, not signal.

The tiers also control the visual complexity of the Time tab's constellation point-clouds (Section 3.6). The Mind and Body clouds grow denser and more articulated as the user's tenure increases, reflecting the accumulation of data and experience. A user with 3 days of dose history sees a sparse field; a user with 3 years sees a rich, structured visualization. The change is gradual enough to be subliminal—most users will not consciously notice the cloud density—but the effect is that the Time tab deepens with use rather than stagnating.

2.3 Honest Uncertainty

The PK model's predictions are population estimates. The companion paper [1] quantifies this: out-of-sample validation errors of 1–15% for estradiol and testosterone single-dose and steady-state checks, with specific structural limitations (TU peak/trough ratio, transdermal intraday ripple, oral testosterone peak fluctuation) that bound what the model can honestly claim.

The interface reflects this uncertainty in five ways:

1. The level overlay on the home screen reads "Estimated" before the numeric value—a one-word qualifier that appears on every readout, establishing that the number is a model output, not a measurement.
2. The calibration factor α is displayed in Settings after a blood test is entered, making the personalization explicit rather than implicit. The user can see that the model has been adjusted, and by how much.
3. The PK model's known limitations—TU peak/trough ratio, transdermal intraday ripple, TP's uncalibrated $t_{\max}$—are *not surfaced in the consumer interface.* This is itself a design decision:

the app does not burden the user with caveats about every prediction. The limitations are documented in the companion paper and in code comments; they constrain what the app claims, but they are not presented as consumer-facing disclaimers. This choice is discussed further in Section 6.8.

4. The therapeutic target bands on the graph are only rendered when the user has configured them. A population-default band could be drawn automatically—the model knows the typical ranges—but doing so would present a clinical reference that the user did not request, blurring the line between information and advice.
5. The graph's visible time window and sampling resolution are derived from the ester's pharmacokinetic properties (Equations 10–11 in [1]), not hardcoded. This means the curve always shows a clinically appropriate window for the specific medication, but it also means the window changes when the medication changes—a subtle consistency trade-off that privileges accuracy over familiarity.

2.4 Per-User Personalization

Between-subject variability in hormone metabolism is substantial, and the app addresses it through a chain of personalization layers, each narrower than the last.

Onboarding configuration. The medication type (estrogen vs. testosterone), ester, dose, concentration, and frequency are the coarsest layer. They select which PK constants to use and anchor the population prediction to the user's reported regimen. These choices are high-confidence: they come directly from the user's prescription.

Blood test calibration. The calibration factor $\alpha = C_{\text{measured}}/C_{\text{predicted}}$ (Section 3.4 of [1]) is a single scalar multiplier on the entire model output, derived from the user's own blood test results. The interface for entering a blood test is minimal: date, measured level, hormone type. The app computes α silently and applies it to all subsequent predictions. The user does not need to understand the factor to benefit from it; the curve simply becomes more accurate after a blood test is recorded.

Body type. The 3D body model used for injection-site tracking offers three silhouettes (neutral, feminine, masculine). This is a visual personalization that affects only the body model's mesh; it does not alter the PK predictions. Its presence in onboarding—step 8 of 11—signals that the app acknowledges anatomical diversity without making it medically consequential.

Therapeutic targets. Users can configure trough minimum, trough maximum, and peak ceiling values in Settings. These numbers draw the reference bands on the home screen graph and gate the comparative text in the level overlay. They are the only user-configurable clinical parameters in the app, and they are explicitly optional: if not set, the app reports estimates without comparison.

Injection zones. Users can mark body zones as "preferred" or "avoid" on the 3D model. The app uses these preferences to suggest rotation patterns, but—critically—it does not enforce them. The user can always tap anywhere on the body; the zone preferences are advisory, not restrictive.

2.5 Accessibility as First Principle

xT's motion and typography systems are designed with accessibility constraints as defaults, not as overrides to be applied later.

Reduce Motion. Every animation in the app is gated on `@Environment(\.accessibilityReduceMotion)`. The `EntranceModifier` (Section 4.2) checks Reduce Motion on appear; when enabled, views appear immediately at their final position with no animation. The confetti burst on dose recording is suppressed. The constellation point-clouds on the Time tab freeze into static arrangements. The spring-driven button press animation degrades to an instant opacity change. The design goal is that no information is ever conveyed *only* through motion; motion is decoration, and decoration must be removable.

Dynamic Type. The app uses role-based font tokens rather than raw `.system(size:)` literals. The tokens map to Apple's built-in Dynamic Type text styles (`.title2`, `.body`, `.caption`, etc.), which scale automatically with the user's system font size preference. Custom tokens (`Theme.Fonts.statValue`, `Theme.Fonts.hero`, `Theme.Fonts.readout`) are built on top of these styles with weight and design adjustments, preserving the scaling behavior. The `@ScaledMetric` property wrapper is used for layout-sensitive values like horizontal padding, so that spacing grows proportionally with text.

Dark-mode-first design. Every color in the design system is defined via `Color(lightHex:, darkHex:)`, with the dark variant treated as the primary and the light variant as the adaptation. The brand gradient, graph background, glass surfaces, and semantic colors all have explicit light-mode counterparts. The app does not rely on Apple's automatic color inversion; every color pair is authored.

VoiceOver. The app uses standard SwiftUI accessibility modifiers (`.accessibilityLabel`, `.accessibilityHint`, `.accessibilityValue`) on interactive elements. The 3D body model presents an accessibility description of selected zones. The graph, however, is not fully navigable by VoiceOver—a limitation discussed in Section 6.6.

3 The User Journey

This section traces the complete user experience from first launch through ongoing mastery. Each subsection describes what the user sees and does, followed by the design rationale.

3.1 Onboarding: 11 Steps to Configuration

The onboarding wizard (`OnboardingView.swift`, approximately 1,600 lines) is the user's first interaction with xT. It is a linear, non-branching sequence of 11 screens (10 for oral routes, which skip the injection-site step), each asking exactly one question. A progress bar at the top shows position within the sequence. Steps fade in and out with a 0.3-second opacity transition; there is no lateral swiping or skipping ahead.

The wizard is presented as a full-screen cover over the main interface. Until it completes, the main interface does not exist from the user's perspective. This is a deliberate commitment: onboarding is not a tutorial that can be dismissed and returned to later; it is the configuration mechanism for the PK engine, and the app cannot function without it.

3.1.1 Step 0: Welcome

A single screen with the word "xT" in 64pt light weight and the subtitle "Hormone therapy tracking." A single button reads "Begin." There is no account creation, no email collection, no permission request. The user is not asked to trust the app before they have seen what it does.

3.1.2 Step 1: Medication Type

"What hormone are you taking?" Two cards: "Estrogen-based HRT" (estradiol, feminizing) and "Testosterone-based HRT" (masculinizing). Each card includes a one-line plain-language description. Selection determines which branch of the PK model will be used and which esters, units, and therapeutic ranges will be shown in subsequent steps.

3.1.3 Step 2: Experience

"How long have you been on HRT?" Two options: "Just starting" or "I've been on it." If the latter, year and month steppers appear (0–20 years, 1–11 months). This value seeds the endogenous baseline model (Section 3.10 of [1]) and the Time tab's tenure tier. It is the only onboarding step

that has no visual consequence in the immediate post-onboarding experience; its effect unfolds over months.

3.1.4 Step 3: Method and Ester

"How do you take it?" Three method cards: Injectable, Cream, Oral. If Injectable is selected, an ester picker appears below with an animated expand transition. Each ester card shows the ester name, its abbreviation (EV, EC, TC, TE, TU, TP), its terminal half-life in days, and its typical time-to-peak. These are the same constants that appear in Table 2 of [1], translated into plain language: "Peak: ∼2 days" rather than "$t_{\max} = 2.10\,\mathrm{d}$."

Why show half-life and peak at onboarding? Because the ester determines the dosing interval, and the dosing interval determines the user's lived experience. A user choosing between testosterone cypionate (peak ∼2 days) and testosterone undecanoate (peak ∼7 days) is making a decision about how frequently they will inject, and the interface gives them the information to make that decision without leaving the app to search.

A frequency stepper ("Every N days," 1–30) appears below the method selection. The default is 7 days for injectables, reflecting the most common clinical practice.

3.1.5 Step 4: Concentration

"What's your vial concentration?" For each ester, a hardcoded list of common concentrations is presented as selectable chips: 40 mg/mL for estradiol valerate, 200 mg/mL for testosterone cypionate, 5 mg/mL for estradiol cypionate. The most common concentration for the selected ester is marked "· most common." A "Custom" option opens a text field for arbitrary values.

This step exists because concentration is *not* derivable from the ester alone. A user on testosterone cypionate might have a 200 mg/mL vial or a 100 mg/mL vial; the difference is a factor of 2 in the computed volume, and getting it wrong would make the mg/mL conversion (Section 3.1.6) silently incorrect. By making concentration an explicit onboarding step rather than a buried setting, the app ensures the conversion is correctly anchored from the start.

3.1.6 Step 5: Dose and Timing

"What's your dose?" This step introduces the mg/mL toggle, one of the most heavily designed interactions in the app. The problem it solves is specific to injectable medications: prescriptions are written in milligrams (e.g., "4 mg estradiol valerate"), but syringes are marked in milliliters. The conversion depends on the vial concentration (mL = mg/concentration), and getting it wrong means injecting the wrong volume.

The step presents a segmented control with two options: "mg" and "mL." In mg mode, a stepper and text field accept the dose in milligrams; the computed volume in mL is shown below in smaller type ("= 0.10 mL at 40 mg/mL"). In mL mode, the fields accept volume directly and compute the mg amount. Changing the concentration picker updates both values in real time.

A row of quick-dose chips (0.5 mL, 1.0 mL, 1.5 mL in mL mode; 2 mg, 4 mg, 6 mg in mg mode) provides one-tap entry for common doses. The chip values adjust based on the selected medication: estradiol shows lower mg values; testosterone shows higher ones.

A date picker sets the date of the user's most recent dose. The *time* of day is not collected at onboarding—the Quick Dose Sheet (Section 3.3) handles time—but the date is needed to place the first data point on the PK curve. The picker defaults to today, which is correct for users who took their dose on the day they set up the app.

Volume warnings. If the computed injection volume exceeds 3.0 mL (the typical maximum for a single intramuscular injection), a warning appears: "This volume is unusually large for a single injection. Check your dose or concentration." This is a safety net, not a blocker—the user can proceed with any value. The warning exists because volume errors are among the most

dangerous data-entry mistakes: a user who accidentally enters their dose in mg when they meant mL (or vice versa) could see wildly incorrect level predictions.

3.1.7 Step 6: Injection Site

"Where did you take it?" (Skipped for oral routes.) This step presents a full-screen, rotatable 3D body model (`BodyModel3DView.swift`, built on SceneKit) with four snap-angle buttons (Front, Back, Left, Right) and free-pan drag support. The user taps on the body surface to mark their injection site. Tapped zones are highlighted with a filled circle; a zone label appears below the model.

The body surface is divided into clinical zones (ventrogluteal, dorsogluteal, vastus lateralis, deltoid, etc.) via a hit-testing engine (`ZoneHitTestEngine.swift`) that maps 3D surface coordinates to zone identifiers. Zones marked as "avoid" in Settings are tinted red and generate a warning on tap. Zones marked as "preferred" are tinted green.

Why a 3D model rather than a 2D diagram or a text list? Because injection site rotation is spatial. Users need to remember *where* on their body they last injected—left ventrogluteal or right, upper quadrant or lower. A flat list of zone names ("Left VG, Right VG, Left DL…") does not convey the spatial relationship; a 3D model does. The user is not asked to learn zone names; they are asked to point at their body.

The step is skippable. A "Skip" button allows the user to proceed without marking a site. This matters because some users may not want to record injection locations, and the app should not block configuration on a preference.

3.1.8 Steps 7–9: Anti-Androgens, Body Type, Profile

These three steps collect supporting information. The anti-androgen step offers four options (None, Spironolactone, Cyproterone/GnRH agonist, Finasteride/Dutasteride) with a one-line medical note for each (e.g., "Spironolactone blocks androgen receptors and may affect estradiol clearance"). The body-type step offers three silhouettes for the 3D model. The profile step collects a name and optional photo.

These steps are placed late in the sequence because they are not required for the PK engine to function. The app can predict levels without knowing about spironolactone (the adjustment is optional) or the user's body type. They are collected at onboarding because they would be unlikely to be discovered in Settings later, and their absence would silently degrade the experience.

3.1.9 Step 10: Review

The final step displays a summary card of all selections—medication, ester, dose, concentration, frequency, last dose date, injection site, anti-androgens, body type, profile name—and a single button: "Begin Tracking." Tapping it calls `commitOnboarding()`, which writes the configuration to the medication manager, logs the first dose record, sets the `hasCompletedOnboarding` flag, and dismisses the onboarding cover to reveal the home screen.

The review step is a commitment moment. The user sees everything they have configured, in one place, before any data is written. There is no "back" button from the home screen to onboarding; the configuration is durable. If the user needs to change something, they do it in Settings, not by re-running onboarding.

3.2 First Empty State

When the home screen first appears after onboarding, the user sees the brand gradient background, the PK curve (empty, with only the initial baseline), the level overlay showing the endogenous baseline (e.g., "Estimated 100 pg/mL"), and the glass card stack at the bottom. If the user skipped dose entry or if the onboarding did not produce a valid configuration, the glass stack is replaced

by an empty state: a pill icon, the text "Set up your medication," and a prompt to add medication details.

The empty state is designed to be aspirational, not discouraging. The graph is visible in the background—the user can see *where* their data will appear—but the foreground message acknowledges that there is nothing to show yet. The action to resolve the empty state (record a dose, configure medication) is always one tap away.

3.3 First Dose: Celebration and Feedback

Recording a dose opens the Quick Dose Sheet (`QuickDoseSheet.swift`), a bottom sheet that slides up from the home screen's action bar. The sheet is organized as a 2-step flow for injectable and topical routes (dose entry → site selection), and a single step for oral routes (dose entry only).

Step 1—Dose entry. The sheet pre-fills today's date and time, the user's configured dose, and the user's concentration. A mg/mL segmented control mirrors the one from onboarding. The dose can be adjusted with a stepper or typed directly. A date/time picker allows recording a dose for a different time (for catch-up entries).

Step 2—Site selection (non-oral routes only). A compact version of the 3D body model appears, pre-rotated to the last-used angle. The user taps to mark the injection site. Below the model, a row of recent sites shows where the last few injections were placed, with the option to re-use a recent site in one tap.

Completion. When the dose is saved, the sheet dismisses and, after a 0.55-second delay (the duration of the settle spring), confetti particles burst from the center of the screen and the level overlay pulses with a celebration glow. The confetti consists of 28 spring-animated particles (circles, rounded rectangles, capsules) in brand colors (rose, lavender, gold). The particles fan outward with randomized velocities, decelerate under simulated gravity, and fade out over 2.5 seconds.

Why confetti? Because dose adherence is the most important behavior the app can encourage, and positive reinforcement at the moment of action is more effective than reminders before the fact. The confetti is celebratory but not gamified—there is no streak counter, no points, no leaderboard. The reward is the moment itself: a brief, beautiful acknowledgment that the user did the thing they came to the app to do.

After the celebration, the PK curve updates. The single dose appears as a rising arc on the graph, peaking at the ester's $t_{\max}$ and then decaying. The level overlay updates to show the current estimated level at the peak position. For a first-time user, this is the moment the app transitions from "I set up a tracker" to "it's tracking me."

3.4 Daily Use: The Home Screen

The home screen (`HomeView.swift`) is the app's centerpiece. Its layout is designed for the 95th-percentile interaction: checking your current level. That interaction requires zero taps, zero scrolling, and less than two seconds of visual processing.

The graph background. `FlatGraphView` renders the PK curve as a full-bleed cubic-bezier path over a dark gradient background with rose and purple ambient glows. The curve's color drifts along a rose-to-lavender ramp keyed to the current level relative to the therapeutic ceiling: low levels render in deeper rose, high levels in lighter lavender. This is ambient information—the user absorbs it peripherally without needing to read numbers. The curve is animated with an `AnimatableVector` that morphs smoothly between data updates, preventing the jarring jumps that would occur if the path were recomputed from scratch on each new dose.

Grid lines, Y-axis labels, and peak/trough reference lines are rendered at low opacity over the curve. The reference lines are drawn from the user's therapeutic targets if configured, or omitted entirely if not.

The level overlay. `LevelOverlayView` is positioned in the top-left corner, floating over the graph. It displays:

- The word "Estimated" in small type
- The current level as a large animated number (e.g., "284"), rendered with `AnimatedCounter` for smooth digit transitions
- The unit ("pg/mL" or "ng/dL")
- The medication name in smaller type
- A status pill showing the level's relationship to the user's therapeutic targets (e.g., "Within set trough")

The overlay is the answer to the question "what is my level right now?"—the most frequent question the app receives. The design prioritizes legibility over density: one number, not a dashboard.

Scrubbing. Touching and dragging anywhere on the graph area activates a scrub mode: the level overlay updates in real time to show the estimated level at the touched time point, the date appears below the level value, and a vertical indicator line marks the position. Lifting the finger returns the overlay to the current-time value. Scrubbing solves the "what was my level on Tuesday?" question without requiring a separate date-picker interface.

The unified glass stack. `UnifiedStackView` is a glass-morphism card (iOS 26 Liquid Glass with `ultraThinMaterial` fallback) containing stacked rows separated by hairline dividers:

1. **Insight row** (`InsightRowView`): Contextual text about the current level estimate. At trough: "Estimated 284 pg/mL at trough—within your set trough range of 100–200 pg/mL." Near peak: "Estimated 612 pg/mL near peak—above your set peak ceiling of 600 pg/mL." If a safety flag is active, an additional note follows with the clinical association. If spironolactone is configured, a clearance note appears: "Spironolactone may increase estradiol clearance by approximately 15%—this estimate accounts for that adjustment."
2. **Mood row** (`MoodRowView`): Today's journal entry, showing the mood emoji, mood name, and a preview of the text. If no entry has been recorded, the row reads "Add mood / Tap to journal" and the entire row is tappable to open the journal composer.
3. **Next dose row** (`NextDoseRowView`): The day name of the next scheduled dose ("Wednesday"), the number of days remaining, and a dot track showing progress through the current cycle. For a 7-day cycle on day 3, three filled dots and four unfilled dots are shown. The dot track is a spatial representation of cycle position that can be read without knowing the numbers.
4. **Action bar** (`ActionBarView`): Two buttons: "Record dose" (primary, rose-tinted) and "Journal" (secondary). These are the two actions the home screen is designed to facilitate.

Each row enters with a staggered spring animation (0.07-second delay between rows, settle spring with 0.55-second duration and zero bounce). The stagger is fast enough to feel like a single cascade rather than a sequence of individual animations. On Reduce Motion, all rows appear immediately.

Why four rows? Each row earns its place by answering a distinct user question: "How am I doing?" (insight), "How do I feel?" (mood), "When's my next dose?" (next dose), "What can I do?" (action bar). Adding more rows would dilute the home screen's primary function as a two-second check; removing any of the four would leave a question unanswered. Four is the number that emerged from iteration, not from a principle; it is the pragmatic answer to "how much information fits before the card becomes a dashboard."

3.5 Calibration: The First Blood Test

Blood test entry lives in Settings, under History → Blood Tests. The composer (`BloodTestComposerView.swift`) collects three fields: date of draw, measured level, and hormone type (estradiol or testosterone). When saved, the app computes:

$$\alpha = \frac{C_{\text{measured}}}{C_{\text{predicted}}}$$

where $C_{\text{predicted}}$ is computed by running the PK model at the exact draw date *without* cal-

ibration applied—the calibration engine never feeds on its own output, preventing circularity. The factor is clamped to $[0.2, 5.0]$ to exclude clinically implausible values that would indicate data-entry error.

The calibration is applied silently. The user does not see α directly unless they view the blood test detail; what they see is that subsequent level estimates on the home screen are closer to their measured values. The Labs Chart view (`LabsChartView.swift`) overlays blood test results on the PK prediction curve, showing the measured values as discrete points and the model prediction as a continuous line. This is the single most direct feedback the app provides about its own accuracy: the user can see, at a glance, how closely the model tracks their actual measurements.

The calibration UX embodies the "honest uncertainty" principle (Section 2.3). The factor is displayed, the model's population prediction is visible alongside the calibrated prediction, and the user can see the magnitude of the adjustment. Nothing is hidden; nothing is oversold.

3.6 The Time Tab: Tenure and Constellations

The Time tab (`TimeView.swift`) is the app's most conceptually ambitious screen. It departs from the home screen's functional density in favor of ambient, exploratory visualization. Its design philosophy is that some information about HRT is better *absorbed* than *read*—that the user's understanding of their own timeline develops through repeated exposure rather than through a single explanatory session.

Constellation point-clouds. The upper portion of the screen contains two large, interactive particle systems: "Mind" (brain-shaped point cloud, indigo palette) and "Body" (silhouette-shaped point cloud, amber palette). Each cloud is composed of hundreds of particles rendered in 3D with subtle drift animation. The clouds are tappable: tapping Mind navigates to mood and cognitive-change content; tapping Body navigates to physical-change content. A body-system strip between them summarizes other systems (Skin, Cycle) with smaller indicator cards.

The cloud density and articulation are driven by the user's months on HRT. A new user sees sparse, loosely-structured clouds; a veteran sees dense, clearly articulated shapes. The progression is continuous, not stepped at tier boundaries—every additional month adds structure. This is ambient personalization: the user is not told "your cloud is denser now," but the Time tab feels richer with use, countering the staleness that affects many health apps' static dashboards.

Tenure tiers. The content model (`TimeContentEngine`) is parameterized by `TenureTier`, a 7-level enumeration: `beginning` (0–2 months), `emerging` (2–6), `inMotion` (6–12), `settled` (12–24), `deepening` (24–60), `established` (60–120), and `veteran` (120+). Each tier unlocks specific content about expected changes, surveillance recommendations, and body-system timelines. The tier system is documented in a dedicated design file (`TIME_TIERS_DESIGN.md`).

Tier transitions trigger a celebration overlay—a brief animated acknowledgment that the user has reached a new stage. Unlike the confetti on dose recording, which is immediate and functional, the tier celebration is reflective: it marks accumulated time, not a single action.

"Up Next" drawer. A narrow drawer peeks from the bottom of the screen, showing only an upward chevron. Pulling it upward triggers a staggered cascade: the Mind and Body sections lift, shrink, and blur away (Mind leading, Body following at a greater distance), revealing a list of upcoming milestones and expected changes. The cascade is the most complex animation in the app, and it is designed to feel like opening a door: the content that was ambient becomes focused, and the content that was focused (the constellations) recedes. On Reduce Motion, the cascade degrades to a cross-fade.

3.7 The Journal: Reflection

The Journal tab (`JournalView.swift`) is the app's reflective surface. Where the home screen answers "what is my level" and the Time tab answers "what is happening to my body," the Journal answers "how do I feel about it."

Composer. The journal composer (`JournalComposerView.swift`) presents a 7-mood picker (Happy, Calm, Neutral, Tired, Anxious, Sad, Energetic) rendered as emoji in a horizontal row, a set of common HRT symptom toggles (skin changes, hot flashes, fatigue, headache, energy boost, mood swings, breast tenderness, nausea), and a free-text field. Each entry is timestamped and automatically tagged with the estimated hormone level at the time of writing.

The level tagging is automatic and non-optional. The user does not choose to attach a level reading; the app attaches it because the relationship between subjective experience and hormone level is precisely what the Journal is designed to help the user discover. A user who notices that their "Anxious" entries cluster around trough days is seeing a pattern that the app does not need to point out explicitly.

Timeline. Entries are displayed in a flat timeline grouped by calendar day, newest first. Each day group shows the date, the primary mood emoji and name, any symptom tags as colored pills, and a text preview. Tapping an entry opens a detail sheet. The timeline is designed for scanning, not searching—its purpose is pattern recognition across weeks and months, not retrieval of a specific entry.

3.8 Configuration: Settings

The Settings tab (`SettingsView.swift`) is organized as a scrollable list of sections, each with a header and a set of tappable rows or inline controls. The section order reflects frequency of access: medication configuration appears near the top; data export and developer settings near the bottom.

Medication management. Type, method, ester, and dose schedule can be changed at any time. Changing the medication type or ester re-triggers the PK model with the new constants; historical dose records retain their original ester assignments, so a user who switches from oral to injectable does not see their oral doses reinterpreted with injectable kinetics.

Body zones. Preferred and avoided injection zones are configured on the 3D body model, with the same interface used during onboarding. The zone configuration screens (`PreferredZonesEditorView.swift`, `AvoidZonesEditorView.swift`) are full-screen editors that allow multi-select zone marking. The zone preferences feed into the Quick Dose Sheet's site-selection step, where avoided zones are tinted red and preferred zones green.

Therapeutic targets. Three numeric fields: trough minimum, trough maximum, and peak ceiling. These are the only user-configurable clinical parameters in the app. They draw the reference bands on the home screen graph and gate the comparative text in the level overlay. The fields are accompanied by a note: "Targets are for reference only and do not constitute medical advice."

History. Separate list views for dose history (`DoseHistoryView.swift`), blood test history (`BloodTestHistoryView.swift`), and screening history (`ScreeningHistoryView.swift`). Each is a chronological list with summary cards. The blood test history includes the Labs Chart view, which overlays measured values on the predicted curve—the most direct accuracy feedback the app provides.

Privacy. Face ID / passcode lock with configurable auto-lock timing (immediate, 1 minute, 5 minutes, 15 minutes, 1 hour). The lock is implemented as a full-screen overlay (`AuthenticationView`) that appears when the app backgrounds, gated by `AuthenticationManager`. Health data is sensitive, and the lock is a binary control: either the app is locked or it is not. There is no partial lock that hides some data and shows other data, because partial locking would create a false sense of security while adding interface complexity.

4 Encoded Design Decisions

This section documents the design tokens, patterns, and conventions that ship in the app. It is the UX analog of the PK paper's Encoded Constants table (Table 2 in [1]): a reference for what

is in the application, where it came from, and what confidence attaches to it.

4.1 Color System

Token	Light	Dark
`accent`	#D4537E	#E8709A
`accentSecondary`	#534AB7	#7E74D8
`successBrand`	#3B8C6E	#64DC9E
`glassTint`	#F0ECF8	#0A0814
`levelLow`	#99345A	#C2557E
`levelHigh`	#8875D4	#B0A4EA
Time category palettes (base / surface / border / text)		
Mind	#534AB7 / #EEEDFE / #CECBF6 / #3C3489	#7E74D8 / #1E1C3A / #3D3880 / #ABA7F0
Skin	#0F6E56 / #E1F5EE / #9FE1CB / #0A4F3E	#2DB88E / #0C2420 / #1A5C46 / #7FE0C4
Body	#854F0B / #FAEEDA / #FAC775 / #5C360A	#D4862B / #2E1C05 / #6B3D08 / #EFB25F

Table 1: Core color tokens. Every color in the application is defined via `Color(lightHex:, darkHex:)`, with explicit light and dark variants. No color relies on automatic inversion.

The color system's design constraint is that every color must work on both light and dark backgrounds, since the app uses a dark brand gradient for onboarding and the home screen background but standard system backgrounds for Settings and sheets. The solution is explicit pairing: every color is authored twice, once for each appearance. The cost is maintenance (adding a color means adding two hex values); the benefit is that no screen looks like an afterthought in either mode.

Why rose and lavender? The brand gradient (rose #D4537E to lavender #8875D4) was chosen to be distinctive without being gendered. The colors reference medical association (red/pink → health, purple → pharmacy) without mapping to binary gender coding (blue for testosterone, pink for estrogen). The gradient appears on the onboarding flow and as an ambient brand marker, never as a functional color.

4.2 Motion System

Preset	API	Duration	Bounce	Use
`settle`	`spring(duration:bounce:)`	0.55	0.00	Entrance animations, card reveals
`pop`	`spring(duration:bounce:)`	0.40	0.15	Interactive responses
`gentle`	`spring(duration:bounce:)`	0.50	0.08	Warm transitions
`tap`	`spring(duration:bounce:)`	0.25	0.05	Button press feedback
`burst`	`spring(duration:bounce:)`	0.60	0.28	Celebrations, confetti

Table 2: Named spring presets. All use the WWDC23 `spring(duration:bounce:)` API. The `settle` spring (0.55 s, 0 bounce) is the default entrance curve; its zero-bounce decay feels natural without calling attention to itself.

Entrance animations. The `EntranceModifier` fades and lifts views into place from below (16pt offset) with a configurable delay. Stacked cards use per-index staggering at 0.07-second intervals via `.entrance(index:)`, producing a cascade that reads as a single motion rather than a sequence. The stagger interval was tuned by feel: at 0.05 s the cascade is imperceptible; at 0.10 s it feels slow. 0.07 s sits at the threshold where the cascade is noticeable but not distracting.

All entrance animations are gated on `@Environment(\.accessibilityReduceMotion)`. When Reduce Motion is enabled, views appear immediately at their final position.

Tactile buttons. `PressableButtonStyle` applies a 0.95 scale reduction and 0.92 opacity reduction on press, animated with the `tap` spring (0.25 s, 0.05 bounce), and fires a soft-impact haptic (`UIImpactFeedbackGenerator(style: .soft)`). The combination of visual compression and haptic feedback makes buttons feel physical—a property that matters for the Record Dose button, which is the most frequently tapped control in the app.

Celebration effects. Two effects fire on dose recording: a confetti burst of 28 particles (circles, rounded rectangles, capsules in brand colors) with randomized trajectories, gravity, and fade; and a celebration glow (`CelebrationGlowModifier`) that pulses the level overlay with a PhaseAnimator-driven bloom. The confetti is gated on Reduce Motion; the glow degrades to a simple opacity pulse.

4.3 Typography

Token	Base Style	Role
`statValue`	Large title, rounded design	Home screen level readout
`hero`	Title, light weight	Onboarding welcome
`readout`	Title 2, rounded design	Secondary numeric displays
`eyebrow`	Caption 1, uppercased	Section labels
`pill`	Caption 2, medium weight	Status badges
`detail`	Subheadline	Form field labels
`label`	Body	Primary body text
`numeric`	Title 3, monospaced	Dose entry fields

Table 3: Role-based font tokens. Every token is built on a Dynamic Type text style, so all text scales with the user's system font size preference. Custom tokens add weight and design adjustments on top of the base style.

The typography system's constraint is that fonts must be legible at very large sizes (the home screen level readout is the largest text in the app, and it must work at Accessibility sizes) and at very small sizes (status pills, axis labels). The solution is role-based tokens that abstract away the underlying font metrics. If a token needs adjustment, it is changed in one place; all views that use the token inherit the change.

4.4 Surface Language

Glass morphism. The unified card stack on the home screen uses a glass effect: on iOS 26, `.glassEffect` produces the system Liquid Glass material; on earlier iOS versions, `ultraThinMaterial` with a tinted background provides a visual fallback. The glass card carries rounded corners (`Theme.Radius.lg`) and a subtle border (`Color.white.opacity(0.2)`). The effect is decorative, not functional—the card would work identically on a solid background—but the glass aesthetic serves a purpose: it visually separates the card's information (read, actionable) from the graph behind it (ambient, decorative), reinforcing the home screen's information hierarchy.

Time cards. Content in the Time tab uses opaque pastel cards colored with the category palette's surface color. These cards are tappable navigation targets, not ambient displays, so they use solid backgrounds rather than glass. The distinction is consistent: glass means "this is an overlay on ambient content"; solid means "this is a discrete piece of information."

Dividers. Separators between rows in the unified stack are 0.5pt rectangles at 35% opacity of the border color. Thinner than the system default (`Divider()`), they mark boundaries without dominating the card. The half-point weight is below the threshold where a line reads as a structural element; it reads as a seam.

4.5 Presentation Rules

The file `HormoneLevelPresentation.swift` (approximately 300 lines) is the single point where hormone level data becomes user-facing text. It exports:

- `LevelStatus` enum: 6 cases from `criticallyLow` to `high`
- `SafetyFlag` enum: 4 clinical threshold crossings
- `ModelDeviation` enum: how calibrated prediction compares to population prediction
- `RangeSource` enum: `userSet` or `modelDerived`
- Formatting functions that accept these enums and return localization-ready strings

The design rule is that no other file in the application may contain a hardcoded string that describes a hormone level in comparative or clinical terms. All such strings flow through this file. This centralization exists for three reasons: it keeps the presentation rules auditable (every instance of "below target" can be found in one file); it ensures consistency (two screens describing the same level status will use the same phrasing); and it simplifies localization (the presentation-rules file is the natural target for translation).

4.6 Navigation Architecture

Tab	Icon	Function	Interaction Model
Home	house	Check level, record dose	Glance + action
Journal	book	Record and review moods	Compose + scan
Time	wand.and.stars	Explore timeline	Browse + discover
Settings	gearshape	Configure	Edit + review

Table 4: The 4-tab architecture. Each tab has a distinct interaction model: Home is for quick actions, Journal for composition and review, Time for browsing, Settings for editing. The models are intentionally different—a tab that duplicated Home's glance-and-act pattern would be redundant.

The 4-tab structure is a standard iOS pattern (`TabView` with `.automatic` style). The tabs are not customizable; their order is fixed. The decision to use four tabs rather than three or five was driven by content, not by principle: Home (dashboard), Journal (reflection), and Settings (configuration) were the three non-negotiable screens, and Time (exploration) earned its place as the fourth by covering a user need—"what's happening to me over time"—that none of the other three addressed.

5 Validation: Persona Journeys

Section-level walkthroughs of screens describe the interface as designed. Persona journeys describe the interface as *used*—by specific people, on specific regimens, encountering specific situations. This section adopts the persona-validation approach of the companion PK paper (Section 4.4 of [1]), exercising UX edge cases rather than PK edge cases.

Persona	Scenario	What it tests, and the result
Maya, 24	4 mg EV IM q7d, first-time user, completes full onboarding	**Onboarding completeness.** Maya sees 11 screens (injectable route), each presenting one decision. The ester picker shows EV's half-life (2.6 d) and peak (∼2 days). She selects 40 mg/mL concentration (marked "most common"), enters 4 mg dose (0.10 mL auto-computed), taps her left ventrogluteal zone on the 3D model, and reviews her configuration before committing. First dose is logged; confetti fires; the PK curve shows a single-dose arc peaking at 2.1 d. The level overlay reads "Estimated 307 pg/mL" near peak. Pass.
Devon, 31	100 mg TC q7d, switches to 200 mg q14d at month 6	**Regimen change UX.** Devon opens Settings → Dose Schedule, changes frequency from 7 to 14 days, changes dose from 100 to 200 mg. The PK model recomputes: C_{avg} is preserved, trough falls, peak rises, peak/trough widens from 1.74× to 3.51×. The home screen curve reshapes to show the longer interval. The next-dose countdown updates to 14 dots. Devon does not need to re-enter onboarding or re-log past doses; the configuration change applies forward. Pass.
Priya, 47	2 mg oral E2 BID for 4 months, then switches to 4 mg EV q7d	**Route switch.** Priya changes her medication type (still Estrogen) and method (Injectable) in Settings. The app prompts for ester, concentration, and dose—the fields that differ between routes. Historical oral doses retain their oral kinetics ($F = 0.05$, fast rates); new injectable doses use EV kinetics. The graph shows a continuous curve across the transition. Priya does not need to delete her history to change routes. Pass.
Sam, 19	50 mg/day T gel, misses ∼2 days/week	**Missed-dose handling.** Sam's dose record status is "missed" for the skipped days. The PK model computes the curve without those contributions. C_{avg} drops to 637 ng/dL; on missed days, the level overlay shows a lower value (roughly 20% below average, 63% below normal trough). The next-dose row continues to show "Today" since gel is daily; the dot track shows one dot (daily cycle). Sam sees the consequence of missed doses without the app moralizing about adherence. Pass.

Persona	Scenario	What it tests, and the result
Rowan, 38	4 mg/day compounded E2 cream, custom concentration	**Custom concentration entry.** Rowan selects "Custom" in the concentration step, types their compounded concentration (e.g., 2.5 mg/mL), and proceeds. The dose step computes mL from the custom concentration correctly. The route pseudo-ester system resolves "Cream" to the transdermal estradiol constants. The graph uses the 24-hourly micro-dose model (Equation 4 in [1]). The custom concentration value persists and appears in Settings. Pass.
Kai, 52	237 mg oral TU BID, 10-day interruption (travel, ran out)	**Washout and re-accumulation.** Kai records no doses for 10 days. The PK curve falls to the endogenous floor within ∼1 day (oral TU's fast kinetics: $k_e = 8.305\,\text{day}^{-1}$, $t_{1/2} = 2.0\,\text{h}$). No phantom drug survives the gap. When Kai resumes, the first new dose appears on the curve at the correct date; the re-accumulation to steady state takes ∼2 days. The level overlay reads near zero during the gap, which is clinically accurate. Kai does not need to tell the app about the interruption; the absence of dose records is the signal. Pass.
Jules, 28	100 mg TE IM q7d, first blood test at month 3	**Calibration UX.** Jules enters a blood test result: 680 ng/dL at trough. The app computes $\alpha = 1.20$ (measured 20% above population prediction). The Labs Chart shows the measured point above the predicted curve. Subsequent level estimates are scaled by 1.20; the home screen now reads ∼680 at trough rather than the population prediction of ∼566. The α factor is visible in the blood test detail but not in the home screen overlay. Jules does not need to understand α to benefit from it. Pass.

Persona	Scenario	What it tests, and the result
Nadia, 35	5 mg EC q10d—an interval nobody studied	**Extrapolation outside the calibration envelope.** Nadia configures a custom 10-day frequency. The PK model handles it—the superposition equation (Equation 2 in [1]) is agnostic to interval—but the resulting curve sits outside the range where the EC constants were anchored (steady-state q7d). The trough at q10d is ∼40% below the q7d trough for the same dose. The app does not warn Nadia that her interval is unconventional; it computes the prediction and presents it. The companion PK paper flags this as "structurally sound but inherits every assumption of the fit"; the interface does not surface that caveat. Pass, with caveat.

All eight personas complete their journeys without interface friction. The personas test properties that section-level screen descriptions cannot: the discoverability of settings for regimen changes (Devon), the handling of route switches with mixed-dose histories (Priya), the consequence visualization of missed doses (Sam), custom-value entry paths (Rowan), washout handling during gaps (Kai), the transparency of calibration (Jules), and the behavior of the interface outside its design envelope (Nadia).

6 Discussion: Design Trade-offs and Limitations

This section describes trade-offs that were consciously accepted and limitations that remain unaddressed. It is the UX analog of the companion PK paper's Discussion (Section 5 of [1]), which quantifies structural limitations rather than hiding them.

6.1 Onboarding Length vs. Setup Completeness

The 11-step onboarding is long relative to consumer app norms. Many health apps complete setup in 3–5 screens, accepting that additional configuration will happen later in Settings. xT's position is that later configuration does not happen—users who skip a setting at onboarding rarely discover it in Settings, and the app's predictions silently degrade as a result.

The trade is that onboarding takes 3–5 minutes, which is above the median for consumer app setup, in exchange for the app being fully configured at first launch. The cost is potential drop-off during onboarding; the benefit is that users who complete it have a correctly parameterized PK model from day one. Whether the drop-off rate justifies the completeness is an empirical question the current design cannot answer without analytics the app does not collect.

6.2 The Graph vs. The Number

The home screen privileges the level overlay (one number) over the PK curve (the full time series). This is a deliberate choice: the number answers the most frequent question immediately; the curve answers follow-up questions on demand (via scrubbing). The cost is that users may not discover the curve's information density—the peak/trough pattern, the accumulation across doses, the shape of the decay—because the interface does not push them to look at it.

An alternative design would foreground the curve and derive the number from it, making the time-series the primary visual and the current-value readout secondary. This would better serve

users who think in terms of curves and cycles, at the cost of slower legibility for the glance-and-go interaction. The current design favors the novice; a future iteration might introduce a mode that favors the experienced user.

6.3 Medical Information Without Medical Advice

Section 2.1 describes the principle. This section describes its cost. The app's refusal to advise means that a user who *wants* directive guidance—"your levels are too high, skip your next dose"—will find xT unsatisfying. The app will tell them their level, tell them it exceeds a clinical threshold, and state the association. It will not tell them what to do about it.

This is the correct design for a consumer health application operating without clinical oversight, but it creates a gap between what some users expect from a "health app" (directive, prescriptive) and what xT provides (informative, associative). Bridging that gap without crossing the line into advice is an unsolved design problem; the current interface acknowledges it by erring on the side of reticence.

6.4 Tenure Tiers as Content Gating

The Time tab's tenure-tier system withholds information from early-stage users. A user in month 1 does not see year-3 surveillance content; a user in year 5 does not see month-1 "first changes" content. The gating is intentional: information presented before it is relevant is noise.

But the system assumes a standard progression through HRT that may not match every user's experience. A user who starts, stops, and restarts HRT—or who is on a non-standard protocol—may see content gated behind a tier that does not reflect their clinical reality. The tenure counter (`activeMonthsOnHRT`) counts months since the first dose record; it does not account for gaps. A user who took HRT for 6 months, stopped for 2 years, and restarted would be in the `inMotion` tier (6–12 months) despite having only a few weeks of current experience. This is a known limitation of the tier model; addressing it would require a more sophisticated treatment-interruption model that the current design does not include.

6.5 Calibration as a Single Scalar

The per-user calibration factor α (Section 3.5) is the UX equivalent of the PK paper's "pragmatic IIV analog" (Section 5.5 of [1]). It is a single scalar multiplier on the entire model output. It cannot represent parameter-specific variability: a user with fast clearance but normal volume of distribution, or a user with high bioavailability but slow absorption, both get the same one-number adjustment.

The UX consequence is that calibration improves accuracy on average but may not improve it at every point in the cycle. A user whose α was derived from a trough measurement may find that their peak predictions are *less* accurate after calibration, because the factor that corrects the trough over-corrects the peak. The interface does not surface this possibility; it shows the calibrated curve as a single line, not as a band that widens to reflect parameter uncertainty. This is a conscious simplification: showing uncertainty bands would be more honest but also more confusing for users who are not accustomed to interpreting confidence intervals.

6.6 What Remains Unaddressed

The following limitations are structural—they cannot be addressed by refining the existing interface and would require new features or architectural changes.

No provider sharing. The app has no mechanism for sharing data with a healthcare provider. A user who wants to show their PK curve to their endocrinologist must take a screenshot. A provider-facing view or export format (PDF report, FHIR integration) would close this

gap but introduces privacy and regulatory complexity (HIPAA compliance, data transmission security) that the current architecture does not address.

No multi-medication tracking. The app tracks one medication at a time. A user on both estradiol and progesterone, or testosterone and finasteride, must choose which to track. The PK model could handle multiple medications in principle, but the interface would need substantial restructuring: a second level overlay, a second curve or a combined curve, and a medication switcher. The current single-medication design is simpler but excludes combination therapy users.

No Apple Health integration. The app does not read from or write to HealthKit. Blood test results, dose records, and journal entries live only within xT. Integration with Apple Health would allow the app to pull lab results entered in other apps and push dose records for use by other health tools. The barrier is not technical but design-level: Apple Health's data model does not have native types for "intramuscular injection of estradiol valerate at 40 mg/mL concentration," so mapping xT's rich dose records to HealthKit's simpler types would lose information.

No social or community features. The app is intentionally single-user. There is no mechanism for comparing experiences, sharing insights, or connecting with other users. This is a product decision, not an oversight: community features introduce moderation burden, privacy risk, and the potential for peer advice to substitute for clinical guidance. But it means the app does not address the social dimension of HRT—the desire to know that other people are experiencing similar things—which is a significant part of the user need.

Limited gender model. The app's medication picker offers two paths: estrogen-based (feminizing) and testosterone-based (masculinizing). This binary framing reflects the current clinical landscape of gender-affirming HRT but does not accommodate non-binary users who may be on regimens that do not map cleanly to either path, or who may prefer not to characterize their HRT in gendered terms. The clinical constants are correct regardless of the label, but the onboarding language could be more inclusive.

English-only. The app is monolingual. The presentation-rules file (`HormoneLevelPresentation.swift`) was designed for localization—classifications are enums, display strings are centralized—and a localization pass is in progress on the current development branch, but as of this writing, every user-facing string is English. For an app targeting a global user base, this is the most significant accessibility gap.

Graph accessibility. The PK curve is not navigable by VoiceOver. A VoiceOver user can hear the current level estimate (the overlay is a standard accessibility element) but cannot explore the curve by scrubbing or hear a description of its shape. Making a continuous mathematical curve accessible to screen-reader navigation is an unsolved problem in iOS accessibility; the app inherits that limitation.

6.7 Design Decision Confidence Hierarchy

Analogous to the PK paper's source quality hierarchy (Section 5.3 of [1]), design decisions in xT rest on a hierarchy of confidence:

1. **Platform-standard** (highest confidence). Decisions that follow iOS Human Interface Guidelines: tab bar navigation, modal sheet presentation, standard gesture recognizers, Dynamic Type, Reduce Motion, VoiceOver labeling. These patterns are tested across millions of apps and billions of user sessions.
2. **Clinically-grounded.** Decisions that follow from the PK model or from published clinical data: the choice of esters, the therapeutic range defaults, the safety flag thresholds, the tenure tier boundaries. These are anchored to the same sources as the PK paper's constants.
3. **Reasoned but untested.** Decisions based on design reasoning without formal user testing: the 11-step onboarding length, the 4-row card stack composition, the mg/mL toggle design, the confetti celebration. These are internally consistent and follow from the design principles, but their effectiveness has not been measured.
4. **Placeholder** (lowest confidence). Decisions known to be suboptimal but retained because a

better alternative is not yet available: TP's uncalibrated $t_{\max}$ (not presented to users, but the curve shape is wrong for the first $\sim$24 hours), the single-α calibration (correct on average, not at every cycle point), the lack of treatment-interruption modeling for tenure tiers.

6.8 Limitation Surfacing

A recurring question in the design of xT is: *which* of the app's known limitations should be surfaced to the user, and *how*? The PK paper documents structural limitations in detail (Sections 5.1–5.5 of [1]), but the consumer interface does not present these caveats alongside predictions. The decision not to surface every limitation is itself a UX choice, and it is worth explaining.

The principle is: surface limitations that the user can act on; document limitations that they cannot. The calibration factor α is surfaced because the user provided the blood test and can see the adjustment. Therapeutic target bands are surfaced because the user configured them and can change them. Safety flags are surfaced because they describe actionable clinical associations.

The PK model's structural limitations—TU peak/trough ratio, transdermal ripple, oral testosterone peak under-prediction, TP uncalibration—are *not* surfaced because the user cannot do anything with that information. Telling a testosterone undecanoate user that "your peak/trough ratio may differ from the model's prediction due to a conflict between published sources" is technically honest but practically useless: it adds cognitive burden without enabling action.

This is a judgment call, and reasonable designers could disagree. The counterargument is that transparency about model limitations builds trust and allows users to calibrate their own confidence in the predictions. The current design accepts the trade: it prioritizes a clean interface over exhaustive caveats, on the reasoning that the user who wants to know about PK model limitations can read the companion paper, and the user who does not should not have caveats forced into their daily experience.

7 Conclusion

We have presented the user experience architecture of xT, a mobile application for hormone replacement therapy monitoring. The interface rests on three principles—progressive disclosure, honest presentation, and per-user personalization—that operate at every level of the application, from the 11-step onboarding wizard to the ambient constellation visualizations on the Time tab.

The UX architecture's key structural features are:

- A **progressive disclosure framework** operating on three timescales simultaneously: onboarding (minutes), per-tab depth (seconds), and tenure-tier content gating (months to years).
- A **presentation-rules file** (`HormoneLevelPresentation.swift`) that centralizes every instance of clinical language, ensuring that the app states facts rather than giving advice, gates comparisons behind user-set targets, and keeps semantics separate from display for localization.
- A **design-token system** (colors, motion, typography, surfaces) that encodes clinical semantics as visual properties: the graph curve's color ramp tracks the therapeutic range, the Time tab's palette system maps body systems to colors, and the motion presets encode interaction semantics (settle for arrivals, burst for celebrations).
- A **per-user calibration interface** that translates a blood test result into a single multiplicative factor α, absorbing the dominant source of individual-level prediction error without requiring the user to understand population pharmacokinetics.
- An **accessibility architecture** in which Reduce Motion, Dynamic Type, and dark/light adaptivity are first principles rather than retrofits, with every animation, font, and color gated on the appropriate system setting.

The eight persona journeys (Section 5) exercise the architecture across onboarding, regimen changes, route switches, missed doses, custom configurations, treatment gaps, calibration, and extrapolation beyond the design envelope. All eight complete without interface friction.

Design trade-offs are named rather than hidden: the 11-step onboarding is long but complete; the home screen privileges a single number over the full curve; the calibration factor is a single scalar that cannot represent parameter-specific variability; the tenure-tier system assumes continuous treatment; and the model's structural limitations are not surfaced in the consumer interface, on the reasoning that users cannot act on them.

What remains unaddressed—provider sharing, multi-medication tracking, Apple Health integration, community features, a more inclusive gender model, localization, and graph accessibility—is stated plainly.

The most important property of a health-tracking interface, in our view, is the same as the most important property of the pharmacokinetic model that drives it: **honesty about what it can and cannot say.** xT's interface is designed to be honest. It reports estimates as estimates, not as facts. It compares only against user-set targets, not against its own population defaults. It states clinical associations without prescribing actions. It names its limitations in documentation, even when it does not surface them in the interface. This honesty—the explicit marking of what is designed, what is reasoned, and what is placeholder—is not the most visible property of the interface, but it is the one that matters most for an application whose users may make medical decisions based on its output.

This paper describes a consumer tracking application's user experience design. It is not clinical guidance, and the interface design decisions described here should not be interpreted as medical recommendations. The application's predictions are population estimates that should not be used to make dosing decisions without laboratory confirmation.

References

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